Supplement: Team metadata

Team	Model	Authors	Methods	Website	Metadata
BIOCOMSC	BIOCOMSC-Gompertz	Martí Català, Enric Álvarez, Sergio Alonso, Daniel López, Clara Prats	Empirical model based on cases and deaths dynamics.	https://biocomsc.upc.edu/en/covid-19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/BIOCOMSC-Gompertz.yml
University of Cologne Covid Metrics	CovidMetrics-epiBATS	Tom Zimmermann, Arne Rodloff	Forecasts are based on TBATS - models (DeLivera, Hyndman and Snyder (2011)) and are updated daily for each German state.	https://tomz.shinyapps.io/coronaLandkreise/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/CovidMetrics-epiBATS.yml
Epiforecasts / London School of Hygiene and Tropical Medicine	epiforecasts-EpiNow2	Nikos Bosse, Sam Abbott, Sebastian Funk	Semi-mechanistic estimation of the time-varying reproduction number for latent infections mapped to reported cases/deaths.	https://epiforecasts.io/EpiNow2	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiforecasts-EpiNow2.yml
epiforecasts	epiforecasts-weeklygrowth	Sam Abbott	A Bayesian autoregressive model using weekly incidence data, application of the forecast.vocs R package.	https://samabbott.co.uk	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiforecasts-weeklygrowth.yml
epiMOX	epiMOX-SUIHTER	Giovanni Ardenghi, Giovanni Ziarelli, Luca Dede', Nicola Parolini, Alfio Quarteroni	Compartmental model SUIHTER	https://www.epimox.polimi.it	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/epiMOX-SUIHTER.yml
European COVID-19 Forecast Hub	EuroCOVIDhub-ensemble	Katharine Sherratt, Nikos Bosse, Sebastian Funk	An ensemble, or model average, of submitted forecasts to the European COVID-19 Forecast Hub.	https://covid19forecasthub.eu/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/EuroCOVIDhub-ensemble.yml
Frankfurt Institute for Advanced Studies & Forschungszentrum Jülich	FIAS_FZJ-Epi1Ger	Maria V. Barbarossa, Jan Fuhrmann, Stefan Krieg, Jan H. Meinke	An extended SEIR model with additional compartments for undetected cases	https://www.fz-juelich.de/SharedDocs/Meldungen/IAS/JSC/DE/2021/2021-01-covid-19.html;jsessionid=F4D5FB4027E871A6F4C2FCAF0F08FC35	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/FIAS_FZJ-Epi1Ger.yml

Supplement: Team metadata

Helmholtz Zentrum fuer Infektionsforschung	HZI-AgeExtendedSEIR	Isti Rodiah, Berit Lange, Pratizio Vanella, Alexander Kuhlmann, Wolfgang Bock	Deterministic SEIR type model	https://www.helmholtz-hzi.de/en/nc/research/research-topics/bacterial-and-viral-pathogens/epidemiology/team/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/HZI-AgeExtendedSEIR.yml
ICM / University of Warsaw	ICM-agentModel	Rafał Bartczuk, Łukasz Górski, Magdalena Gruziel-Słomka, Artur Kaczorek, Jan Kisielewski, Antoni Moszyński, Karol Niedzielewski, Jędrzej Nowosielski, Maciej Radwan, Franciszek Rakowski, Marcin Semeniuk, Jakub Zieliński	Agent-based model	https://covid-19.icm.edu.pl/en/model-description/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ICM-agentModel.yml
IEM Health	IEM_Health-CovidProject	Brad Suchoski, Steve Stage, Heidi Gurung, Sid Baccam	SEIR model projections for daily incident confirmed COVID cases and deaths by using AI to fit actual cases observed.	https://iem.com/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/IEM_Health-CovidProject.yml
ILM	ILM-EKF	Stefan Heyder, Thomas Hotz	Extended Kalman filter based on reproduction equation	https://github.com/Stochastik-TU-Ilmenau	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ILM-EKF.yml
Fraunhofer Institute for Industrial Mathematics ITWM	itwm-dSEIR	Jan Mohring, Neele Leithäuser, Michael Helmling	Integral equation model based on age cohorts taking into account vaccination and testing. The parameters are adjusted to the counted cases and deaths.	https://www.itwm.fraunhofer.de/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/itwm-dSEIR.yml
ITWW	ITWW-county_repro	Przemyslaw Biecek, Viktor Bezborodov, Marcin Bodych, Jan Pablo Burgard, Stefan Heyder, Thomas Hotz, Tyll Krüger	Forecasts of county level incidence based on regional reproduction numbers.	https://github.com/Stochastik-TU-Ilmenau	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ITWW-county_repro.yml
JBUD	JBUD-HMXK	Jozef Budzinski	Heavily modified infection-age SIR-X model with waning immunity, vaccinations, seasonality and undetected cases.	https://joebud.github.io/covid-19-analyses/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/JBUD-HMXK.yml

Supplement: Team metadata

MOCOS group	MOCOS-agent1	Marek Bawiec, Marcin Bodych, Tyll Krueger, Tomasz Ozanski, Barbara Pabjan, Agata Migalska, Przemyslaw Biecek, Viktor Bezborodov, Ewa Szczurek, Ewaryst Rafajłowicz, Ewa Rafajłowicz, Wojciech Rafajłowicz	Agent-based microsimulation model	https://mocos.pl/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MOCOS-agent1.yml
Masaryk University	MUNI-ARIMA	Andrea Kraus, David Kraus	ARIMA model with outlier detection fitted to transformed weekly aggregated series.	https://krausstat.shinyapps.io/covid19global/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MUNI-ARIMA.yml
Department of Mathematics and Statistics Masaryk University Team	MUNI_DMS-SEIAR	Veronika Eclerova, Lenka Pribylova	SEIAR model with A compartment of absent unobserved infected estimated from hospital data with incorporated mobility data dependence; optimized to the compartment of all exposed (unobserved included)	https://webstudio.shinyapps.io/MAMES/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/MUNI_DMS-SEIAR.yml
Grzegorz Redlarski	PL_GRedlarski-DistrictsSum	Grzegorz Redlarski	Modified SIR method, applied to all districts. Forecasts for districts are summed up.	https://docs.google.com/spreadsheets/d/e/2PACX-1vRpH4yhKRts7Co5tydhZhojIPTcTTybms1PqJ9j1tmSBzzPLoU2U9XjUWDwiKYxnE6gMLayl71rpGC8/pubhtml?gid=493251550&single=true	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/PL_GRedlarski-DistrictsSum.yml
prolix	prolix-euclidean	Loïc Pottier	Offsets obtained by correlations, best linear approximation of reproduction rates (using vaccination approximation) by least euclidean distance, and linear prediction.	https://cp.lpmib.fr/medias/covid19/_synthese.html	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/prolix-euclidean.yml
Robert Walraven	RobertWalraven-ESG	Robert Walraven	Multiple skewed gaussian distribution peaks fit to raw data	http://rwalraven.com/COVID19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/RobertWalraven-ESG.yml

Supplement: Team metadata

Swiss Data Science Center / University of Geneva	SDSC_ISG-TrendModel	Ekaterina Krymova, Dorina Thanou, Benjamin Bejar Haro, Tao Sun, Gavin Lee, Elisa Manetti, Christine Choirat, Antoine Flahault, Guillaume Obozinski	The Trend Model predicts daily cases and deaths using linear extrapolation on the linear or log scale of the underlying trend estimated by a robust LOESS seasonal-trend decomposition model.	https://renkulab.shinyapps.io/COVID-19-Epidemic-Forecasting/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/SDSC_ISG-TrendModel.yml
Statgroup19	Statgroup19-richards	Pierfrancesco Alaimo Di Loro, Fabio Divino, Alessio Farcomeni, Giovanna Jona Lasinio, Antonello Maruotti, Marco Mingione, Gianfranco Lovison	Richards' curve based generalized growth model	https://statgroup19.shinyapps.io/Covid19App/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/Statgroup19-richards.yml
Statgroup19	Statgroup19-spatialrichards	Pierfrancesco Alaimo Di Loro, Fabio Divino, Alessio Farcomeni, Giovanna Jona Lasinio, Antonello Maruotti, Marco Mingione, Gianfranco Lovison	Richards' curve based generalized growth model taking into account spatial dependence	https://statgroup19.shinyapps.io/Covid19App/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/Statgroup19-spatialrichards.yml
Universidad Carlos III de Madrid	UC3M-EpiGraph	David E. Singh, Miguel Guzman Merino, Maria Cristina Marinescu, Jesus Carretero, Alberto Cascajo Garcia	Agent-based parallel simulator that models individual interactions extracted from social networks and demographical data.	https://www.arcos.inf.uc3m.es/epigraph/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UC3M-EpiGraph.yml
University of Ljubljana, Faculty of Health Sciences Team	ULZF-SEIRC19SI	Janez Zibert	SEIHR model extended with compartments for hospitals, intensive care units, asymptomatic cases, separate submodels for vaccinated and unvaccinated, divided to 5 age subgroups of population	https://apps.lusy.fri.uni-lj.si	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/ULZF-SEIRC19SI.yml
UMass-Amherst	UMass-MechBayes	Dan Sheldon, Graham Gibson, Nick Reich	Bayesian compartmental model with observations on cumulative case counts and cumulative deaths. Model is fit independently to each state. Model includes observation noise and a case detection rate.	https://github.com/dsheldon/covid	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UMass-MechBayes.yml
UNED	UNED-PreCoV2	José L. Aznarte, César Pérez, José Almagro, Pedro Álvarez, Álvaro Ortiz, Fernando Blat	Bayesian time series models with ARIMA noise and fixed transfer functions for each input.	https://precov2.org	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UNED-PreCoV2.yml

Supplement: Team metadata

University of Perugia / University of Milano-Bicocca / Università della Svizzera Italiana	UpgUmibUsi-MultiBayes	Francesco Bartolucci, Fulvia Pennoni, Antonietta Mira	Bayesian Dirichlet-Multinomial models for counts of patients in mutually exclusive and exhaustive categories such as hospitalized in regular wards and in intensive care units, deceased and recovered	https://github.com/francescobartolucci/ARMultinomial	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UpgUmibUsi-MultiBayes.yml
University of Southern California	USC-SIkJalpha	Ajitesh Srivastava, Frost Tianjian Xu	A heterogeneous infection rate model with human mobility for epidemic modeling. Our model adapts to changing trends and provide predictions of confirmed cases and deaths.	https://scc-usc.github.io/ReCOVER-COVID-19	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/USC-SIkJalpha.yml
University of Virginia, Biocomplexity COVID-19 Response Team	UVA-Ensemble	Aniruddha Adiga, Lijing Wang, Srinivasan Venkatramanan, Akhil Sai Peddireddy, Benjamin Hurt, Przemyslaw Porebski, Bryan Lewis, Madhav Marathe, Jiangzhou Chen, Anil Vullikanti	An ensemble of multiple methods such as auto-regressive (AR)models with exogenous variables, Long short-term memory (lSTM) models,Kalman filter and PatchSim (an SEIR model).	https://biocomplexity.virginia.edu/	https://raw.githubusercontent.com/epiforecasts/covid19-forecast-hub-europe/main/model-metadata/UVA-Ensemble.yml